

1. Настройка коммутаторов:

Назначение корневого коммутатора:

В качестве корневого выберем коммутатор Layer2Switch-1

```
conf t
spanning-tree vlan 1 priority 4096 # Назначение низкого приоритета STP
end
copy running-config startup-config
```

```
vIOS-L2-01>enable
vIOS-L2-01#conf t
Enter configuration commands, one per line. End with CNTL/Z.
vIOS-L2-01(config)#spanning-tree vlan 1 priority 4096
vIOS-L2-01(config)#end
vIOS-L2-01#
*Dec 10 02:48:05.007: %SYS-5-CONFIG_I: Configured from console by console
vIOS-L2-01#copy running-cofig strtup-config
Destination filename [strtup-config]?
%Error opening flash0:running-cofig (File not found)
```

Чтобы проверить:

```
show spaning-tree vlan 1
```

```
vIOS-L2-01#show spanning-tree vlan 1

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    4097
             Address     0c3a.993f.0000
             This bridge is the root
             Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    4097 (priority 4096 sys-id-ext 1)
             Address     0c3a.993f.0000
             Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time 300 sec

Interface Role Sts Cost Prio.Nbr Type
-----
Gi0/0 Desg FWD 4 128.1 Shr
Gi0/1 Desg FWD 4 128.2 Shr
Gi0/2 Desg FWD 4 128.3 Shr
Gi0/3 Desg FWD 4 128.4 Shr
Gi1/0 Desg FWD 4 128.5 Shr
Gi1/1 Desg FWD 4 128.6 Shr
Gi1/2 Desg FWD 4 128.7 Shr
Gi1/3 Desg FWD 4 128.8 Shr
Gi2/0 Desg FWD 4 128.9 Shr
```

Видно, что приоритет поменялся на 4096, а BID стал 4097, а все порты стали DP.

В качестве запасного рута выберем Layer2Switch-2:

```
conf t
spanning-tree vlan 1 priority 8192
end
copy running-config startup-config
```

```
vIOS-L2-01>enable
vIOS-L2-01#conf t
Enter configuration commands, one per line. End with CNTL/Z.
vIOS-L2-01(config)#spanning-tree vlan 1 priority 8192
vIOS-L2-01(config)#end
vIOS-L2-01#c
*Dec 10 04:30:17.285: %SYS-5-CONFIG_I: Configured from console by console
vIOS-L2-01#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
Compressed configuration from 5277 bytes to 2019 bytes[OK]
vIOS-L2-01#
*Dec 10 04:30:44.785: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updated on disk. Please wait...
*Dec 10 04:30:45.554: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk successfully.
```

Проверка:

```
show spanning-tree vlan 1
```

```
vIOS-L2-01>show spanning-tree vlan 1

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    4097
            Address     0c3a.993f.0000
            Cost        4
            Port        1 (GigabitEthernet0/0)
            Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    8193  (priority 8192 sys-id-ext 1)
            Address     0cfe.1166.0000
            Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
            Aging Time   300 sec

Interface                Role Sts Cost      Prio.Nbr Type
-----
Gi0/0                    Root FWD 4        128.1   Shr
Gi0/1                    Altn BLK 4        128.2   Shr
Gi0/2                    Desg FWD 4        128.3   Shr
Gi0/3                    Desg FWD 4        128.4   Shr
Gil/0                    Desg FWD 4        128.5   Shr
Gil/1                    Desg FWD 4        128.6   Shr
Gil/2                    Desg FWD 4        128.7   Shr
--More--
```

2. Проверка доступности ПК

Назначаем IP-адреса всех ПК в сети:

```
ip 192.168.1.n/24    # Для каждого ПК. Вместо n – номер ПК
```

PC1: 192.168.1.1

PC2: 192.168.1.2

PC3: 192.168.1.3

PC4: 192.168.1.4

PC5: 192.168.1.5

PC6: 192.168.1.6

Теперь пингуем все компьютеры со всех компьютеров:

ping_pc1.log

===== PuTTY log 2025.12.10 11:59:25 =====

show ping 192.168.1.2

84 bytes from 192.168.1.2 icmp_seq=1 ttl=64 time=6.597 ms

84 bytes from 192.168.1.2 icmp_seq=2 ttl=64 time=1.484 ms

84 bytes from 192.168.1.2 icmp_seq=3 ttl=64 time=0.794 ms

84 bytes from 192.168.1.2 icmp_seq=4 ttl=64 time=0.838 ms

84 bytes from 192.168.1.2 icmp_seq=5 ttl=64 time=0.855 ms

PC1> ping 192.168.1.2 3

84 bytes from 192.168.1.3 icmp_seq=1 ttl=64 time=8.901 ms

84 bytes from 192.168.1.3 icmp_seq=2 ttl=64 time=5.321 ms

84 bytes from 192.168.1.3 icmp_seq=3 ttl=64 time=8.801 ms

84 bytes from 192.168.1.3 icmp_seq=4 ttl=64 time=1.809 ms

84 bytes from 192.168.1.3 icmp_seq=5 ttl=64 time=5.307 ms

PC1> ping 192.168.1.3 4

84 bytes from 192.168.1.4 icmp_seq=1 ttl=64 time=9.848 ms

84 bytes from 192.168.1.4 icmp_seq=2 ttl=64 time=3.163 ms

84 bytes from 192.168.1.4 icmp_seq=3 ttl=64 time=5.044 ms

84 bytes from 192.168.1.4 icmp_seq=4 ttl=64 time=4.547 ms

84 bytes from 192.168.1.4 icmp_seq=5 ttl=64 time=7.182 ms

PC1> ping 192.168.1.4 5

84 bytes from 192.168.1.5 icmp_seq=1 ttl=64 time=4.410 ms

84 bytes from 192.168.1.5 icmp_seq=2 ttl=64 time=6.334 ms

84 bytes from 192.168.1.5 icmp_seq=3 ttl=64 time=1.930 ms

84 bytes from 192.168.1.5 icmp_seq=4 ttl=64 time=11.597 ms

84 bytes from 192.168.1.5 icmp_seq=5 ttl=64 time=8.052 ms

PC1> ping 192.168.1.5 6

84 bytes from 192.168.1.6 icmp_seq=1 ttl=64 time=15.885 ms

84 bytes from 192.168.1.6 icmp_seq=2 ttl=64 time=6.957 ms

84 bytes from 192.168.1.6 icmp_seq=3 ttl=64 time=3.901 ms

84 bytes from 192.168.1.6 icmp_seq=4 ttl=64 time=4.749 ms

84 bytes from 192.168.1.6 icmp_seq=5 ttl=64 time=5.606 ms

PC1>

ping_pc2.log

===== PuTTY log 2025.12.10 12:02:06 =====

ping 192.168.1.1

84 bytes from 192.168.1.1 icmp_seq=1 ttl=64 time=3.177 ms

84 bytes from 192.168.1.1 icmp_seq=2 ttl=64 time=8.835 ms

84 bytes from 192.168.1.1 icmp_seq=3 ttl=64 time=7.969 ms

84 bytes from 192.168.1.1 icmp_seq=4 ttl=64 time=8.433 ms

84 bytes from 192.168.1.1 icmp_seq=5 ttl=64 time=1.756 ms

PC2> ping 192.168.1.1 3

84 bytes from 192.168.1.3 icmp_seq=1 ttl=64 time=12.744 ms

84 bytes from 192.168.1.3 icmp_seq=2 ttl=64 time=6.832 ms

84 bytes from 192.168.1.3 icmp_seq=3 ttl=64 time=13.183 ms

84 bytes from 192.168.1.3 icmp_seq=4 ttl=64 time=3.026 ms

84 bytes from 192.168.1.3 icmp_seq=5 ttl=64 time=7.874 ms

PC2> ping 192.168.1.3 4

84 bytes from 192.168.1.4 icmp_seq=1 ttl=64 time=4.536 ms

84 bytes from 192.168.1.4 icmp_seq=2 ttl=64 time=8.624 ms

84 bytes from 192.168.1.4 icmp_seq=3 ttl=64 time=13.615 ms

84 bytes from 192.168.1.4 icmp_seq=4 ttl=64 time=8.581 ms

84 bytes from 192.168.1.4 icmp_seq=5 ttl=64 time=8.146 ms

PC2> ping 192.168.1.4 5

84 bytes from 192.168.1.5 icmp_seq=1 ttl=64 time=8.926 ms

84 bytes from 192.168.1.5 icmp_seq=2 ttl=64 time=10.831 ms

84 bytes from 192.168.1.5 icmp_seq=3 ttl=64 time=9.591 ms

84 bytes from 192.168.1.5 icmp_seq=4 ttl=64 time=4.564 ms

84 bytes from 192.168.1.5 icmp_seq=5 ttl=64 time=12.776 ms

PC2> ping 192.168.1.5 6

84 bytes from 192.168.1.6 icmp_seq=1 ttl=64 time=9.250 ms

84 bytes from 192.168.1.6 icmp_seq=2 ttl=64 time=10.842 ms

84 bytes from 192.168.1.6 icmp_seq=3 ttl=64 time=5.008 ms
84 bytes from 192.168.1.6 icmp_seq=4 ttl=64 time=7.595 ms
84 bytes from 192.168.1.6 icmp_seq=5 ttl=64 time=7.458 ms

PC2>

ping_pc3.log

===== PuTTY log 2025.12.10 12:03:18 =====

ping 192.168.1.1

84 bytes from 192.168.1.1 icmp_seq=1 ttl=64 time=15.821 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=64 time=16.134 ms
84 bytes from 192.168.1.1 icmp_seq=3 ttl=64 time=7.298 ms
84 bytes from 192.168.1.1 icmp_seq=4 ttl=64 time=7.303 ms
84 bytes from 192.168.1.1 icmp_seq=5 ttl=64 time=3.780 ms

PC3> ping 192.168.1.1 2

84 bytes from 192.168.1.2 icmp_seq=1 ttl=64 time=14.201 ms
84 bytes from 192.168.1.2 icmp_seq=2 ttl=64 time=4.697 ms
84 bytes from 192.168.1.2 icmp_seq=3 ttl=64 time=5.575 ms
84 bytes from 192.168.1.2 icmp_seq=4 ttl=64 time=8.169 ms
84 bytes from 192.168.1.2 icmp_seq=5 ttl=64 time=7.869 ms

PC3> ping 192.168.1.2 4

84 bytes from 192.168.1.4 icmp_seq=1 ttl=64 time=7.610 ms
84 bytes from 192.168.1.4 icmp_seq=2 ttl=64 time=3.050 ms
84 bytes from 192.168.1.4 icmp_seq=3 ttl=64 time=4.584 ms
84 bytes from 192.168.1.4 icmp_seq=4 ttl=64 time=0.807 ms
84 bytes from 192.168.1.4 icmp_seq=5 ttl=64 time=0.742 ms

PC3> ping 192.168.1.4 5

84 bytes from 192.168.1.5 icmp_seq=1 ttl=64 time=3.476 ms
84 bytes from 192.168.1.5 icmp_seq=2 ttl=64 time=7.999 ms
84 bytes from 192.168.1.5 icmp_seq=3 ttl=64 time=1.991 ms
84 bytes from 192.168.1.5 icmp_seq=4 ttl=64 time=16.857 ms
84 bytes from 192.168.1.5 icmp_seq=5 ttl=64 time=11.070 ms

PC3> ping 192.168.1.5 6

```
84 bytes from 192.168.1.6 icmp_seq=1 ttl=64 time=15.336 ms
84 bytes from 192.168.1.6 icmp_seq=2 ttl=64 time=9.430 ms
84 bytes from 192.168.1.6 icmp_seq=3 ttl=64 time=5.633 ms
84 bytes from 192.168.1.6 icmp_seq=4 ttl=64 time=4.273 ms
84 bytes from 192.168.1.6 icmp_seq=5 ttl=64 time=7.150 ms
```

```
PC3>
```

```
ping_pc4.log
```

```
=~=~=~=~=~=~=~=~=~=~=~=~= PuTTY log 2025.12.10 12:04:37 ~=~=~=~=~=~=~=~=~=~=~=~=
```

```
ping 192.168.1.1
```

```
84 bytes from 192.168.1.1 icmp_seq=1 ttl=64 time=11.863 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=64 time=8.111 ms
84 bytes from 192.168.1.1 icmp_seq=3 ttl=64 time=10.412 ms
84 bytes from 192.168.1.1 icmp_seq=4 ttl=64 time=1.748 ms
84 bytes from 192.168.1.1 icmp_seq=5 ttl=64 time=8.749 ms
```

```
PC4> ping 192.168.1.1 2
```

```
84 bytes from 192.168.1.2 icmp_seq=1 ttl=64 time=7.865 ms
84 bytes from 192.168.1.2 icmp_seq=2 ttl=64 time=12.383 ms
84 bytes from 192.168.1.2 icmp_seq=3 ttl=64 time=4.583 ms
84 bytes from 192.168.1.2 icmp_seq=4 ttl=64 time=3.994 ms
84 bytes from 192.168.1.2 icmp_seq=5 ttl=64 time=10.421 ms
```

```
PC4> ping 192.168.1.2 3
```

```
84 bytes from 192.168.1.3 icmp_seq=1 ttl=64 time=5.040 ms
84 bytes from 192.168.1.3 icmp_seq=2 ttl=64 time=1.427 ms
84 bytes from 192.168.1.3 icmp_seq=3 ttl=64 time=0.667 ms
84 bytes from 192.168.1.3 icmp_seq=4 ttl=64 time=7.827 ms
84 bytes from 192.168.1.3 icmp_seq=5 ttl=64 time=7.101 ms
```

```
PC4> ping 192.168.1.3 5
```

```
84 bytes from 192.168.1.5 icmp_seq=1 ttl=64 time=19.187 ms
84 bytes from 192.168.1.5 icmp_seq=2 ttl=64 time=1.659 ms
84 bytes from 192.168.1.5 icmp_seq=3 ttl=64 time=3.831 ms
84 bytes from 192.168.1.5 icmp_seq=4 ttl=64 time=9.020 ms
84 bytes from 192.168.1.5 icmp_seq=5 ttl=64 time=11.798 ms
```

```
PC4> ping 192.168.1.5 6
```

```
84 bytes from 192.168.1.6 icmp_seq=1 ttl=64 time=13.150 ms
84 bytes from 192.168.1.6 icmp_seq=2 ttl=64 time=10.232 ms
84 bytes from 192.168.1.6 icmp_seq=3 ttl=64 time=9.263 ms
84 bytes from 192.168.1.6 icmp_seq=4 ttl=64 time=6.969 ms
84 bytes from 192.168.1.6 icmp_seq=5 ttl=64 time=4.603 ms
```

```
PC4>
```

```
ping_pc5.log
```

```
=~=~=~=~=~=~=~=~=~=~=~=~= PuTTY log 2025.12.10 12:05:41 ~=~=~=~=~=~=~=~=~=~=~=~=
ping 192.168.1.1
```

```
84 bytes from 192.168.1.1 icmp_seq=1 ttl=64 time=7.844 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=64 time=5.018 ms
84 bytes from 192.168.1.1 icmp_seq=3 ttl=64 time=7.342 ms
84 bytes from 192.168.1.1 icmp_seq=4 ttl=64 time=7.849 ms
84 bytes from 192.168.1.1 icmp_seq=5 ttl=64 time=7.400 ms
```

```
PC5> ping 192.168.1.1 2
```

```
84 bytes from 192.168.1.2 icmp_seq=1 ttl=64 time=9.727 ms
84 bytes from 192.168.1.2 icmp_seq=2 ttl=64 time=6.657 ms
84 bytes from 192.168.1.2 icmp_seq=3 ttl=64 time=12.232 ms
84 bytes from 192.168.1.2 icmp_seq=4 ttl=64 time=4.377 ms
84 bytes from 192.168.1.2 icmp_seq=5 ttl=64 time=12.990 ms
```

```
PC5> ping 192.168.1.2 3
```

```
84 bytes from 192.168.1.3 icmp_seq=1 ttl=64 time=8.441 ms
84 bytes from 192.168.1.3 icmp_seq=2 ttl=64 time=6.754 ms
84 bytes from 192.168.1.3 icmp_seq=3 ttl=64 time=4.566 ms
84 bytes from 192.168.1.3 icmp_seq=4 ttl=64 time=4.627 ms
84 bytes from 192.168.1.3 icmp_seq=5 ttl=64 time=14.954 ms
```

```
PC5> ping 192.168.1.3 4
```

```
84 bytes from 192.168.1.4 icmp_seq=1 ttl=64 time=7.346 ms
84 bytes from 192.168.1.4 icmp_seq=2 ttl=64 time=7.915 ms
84 bytes from 192.168.1.4 icmp_seq=3 ttl=64 time=5.613 ms
84 bytes from 192.168.1.4 icmp_seq=4 ttl=64 time=4.821 ms
84 bytes from 192.168.1.4 icmp_seq=5 ttl=64 time=5.052 ms
```

```
PC5> ping 192.168.1.4 6
```

```
84 bytes from 192.168.1.6 icmp_seq=1 ttl=64 time=2.199 ms
84 bytes from 192.168.1.6 icmp_seq=2 ttl=64 time=0.910 ms
84 bytes from 192.168.1.6 icmp_seq=3 ttl=64 time=6.584 ms
84 bytes from 192.168.1.6 icmp_seq=4 ttl=64 time=0.793 ms
84 bytes from 192.168.1.6 icmp_seq=5 ttl=64 time=0.756 ms
```

```
PC5>
```

```
ping_pc6.log
```

```
=~=~=~=~=~=~=~=~=~=~=~=~= PuTTY log 2025.12.10 12:06:50 ~=~=~=~=~=~=~=~=~=~=~=~=
```

```
pinh 192.168.1.1
```

```
Bad command: "pinh 192.168.1.1". Use ? for help.
```

```
PC6> pinh 192.168.1.1inh 192.168.1.1 g 192.168.1.1
```

```
84 bytes from 192.168.1.1 icmp_seq=1 ttl=64 time=13.954 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=64 time=7.909 ms
84 bytes from 192.168.1.1 icmp_seq=3 ttl=64 time=11.022 ms
84 bytes from 192.168.1.1 icmp_seq=4 ttl=64 time=7.157 ms
84 bytes from 192.168.1.1 icmp_seq=5 ttl=64 time=10.625 ms
```

```
PC6> ping 192.168.1.1 2
```

```
84 bytes from 192.168.1.2 icmp_seq=1 ttl=64 time=12.899 ms
84 bytes from 192.168.1.2 icmp_seq=2 ttl=64 time=7.422 ms
84 bytes from 192.168.1.2 icmp_seq=3 ttl=64 time=8.400 ms
84 bytes from 192.168.1.2 icmp_seq=4 ttl=64 time=18.690 ms
84 bytes from 192.168.1.2 icmp_seq=5 ttl=64 time=9.230 ms
```

```
PC6> ping 192.168.1.2 3
```

```
84 bytes from 192.168.1.3 icmp_seq=1 ttl=64 time=15.973 ms
84 bytes from 192.168.1.3 icmp_seq=2 ttl=64 time=6.190 ms
84 bytes from 192.168.1.3 icmp_seq=3 ttl=64 time=7.408 ms
84 bytes from 192.168.1.3 icmp_seq=4 ttl=64 time=9.309 ms
84 bytes from 192.168.1.3 icmp_seq=5 ttl=64 time=5.883 ms
```

```
PC6> ping 192.168.1.3 4
```

```
84 bytes from 192.168.1.4 icmp_seq=1 ttl=64 time=11.591 ms
84 bytes from 192.168.1.4 icmp_seq=2 ttl=64 time=7.656 ms
84 bytes from 192.168.1.4 icmp_seq=3 ttl=64 time=5.021 ms
```


84 bytes from 192.168.1.4 icmp_seq=4 ttl=64 time=8.132 ms

84 bytes from 192.168.1.4 icmp_seq=5 ttl=64 time=4.658 ms

PC6> ping 192.168.1.4 5

84 bytes from 192.168.1.5 icmp_seq=1 ttl=64 time=6.699 ms

84 bytes from 192.168.1.5 icmp_seq=2 ttl=64 time=7.228 ms

84 bytes from 192.168.1.5 icmp_seq=3 ttl=64 time=0.621 ms

84 bytes from 192.168.1.5 icmp_seq=4 ttl=64 time=7.275 ms

84 bytes from 192.168.1.5 icmp_seq=5 ttl=64 time=7.684 ms

PC6>

Каждый компьютер пингуется с каждого компьютера.

3. Обозначение портов

Для вывода нужной информации, нужно ввести в консоль каждого коммутатора:

```
show spanning-tree vlan 1    # Для вывода информации о портах самого коммутатора
```

Layer2Switch-1:

```
vIOS-L2-01>show spanning-tree vlan 1
```

VLAN0001

Spanning tree enabled protocol ieee

Root ID Priority 4097
Address 0c3a.993f.0000
This bridge is the root
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 4097 (priority 4096 sys-id-ext 1)
Address 0c3a.993f.0000
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.Nbr	Type
Gi0/0	Desg	FWD	4	128.1	Shr
Gi0/1	Desg	FWD	4	128.2	Shr
Gi0/2	Desg	FWD	4	128.3	Shr
Gi0/3	Desg	FWD	4	128.4	Shr
Gi1/0	Desg	FWD	4	128.5	Shr
Gi1/1	Desg	FWD	4	128.6	Shr
Gi1/2	Desg	FWD	4	128.7	Shr
Gi1/3	Desg	FWD	4	128.8	Shr
Gi2/0	Desg	FWD	4	128.9	Shr

Layer2Switch-2:

```
vIOS-L2-01>show spanning-tree vlan 1
```

```
VLAN0001
```

```
Spanning tree enabled protocol ieee
```

```
Root ID    Priority    4097
           Address    0c3a.993f.0000
           Cost      4
           Port      1 (GigabitEthernet0/0)
           Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
```

```
Bridge ID  Priority    8193 (priority 8192 sys-id-ext 1)
           Address    0cfe.1166.0000
           Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
           Aging Time 300 sec
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Gi0/0	Root	FWD	4	128.1	Shr
Gi0/1	Altn	BLK	4	128.2	Shr
Gi0/2	Desg	FWD	4	128.3	Shr
Gi0/3	Desg	FWD	4	128.4	Shr
Gi1/0	Desg	FWD	4	128.5	Shr
Gi1/1	Desg	FWD	4	128.6	Shr
Gi1/2	Desg	FWD	4	128.7	Shr
Gi1/3	Desg	FWD	4	128.8	Shr
Gi2/0	Desg	FWD	4	128.9	Shr

Layer2Switch-3:

```
vIOS-L2-01>show spanning-tree vlan 1
```

```
VLAN0001
```

```
Spanning tree enabled protocol ieee
```

```
Root ID    Priority    4097
           Address    0c3a.993f.0000
           Cost      4
           Port      1 (GigabitEthernet0/0)
           Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
```

```
Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
           Address    0c22.03c8.0000
           Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
           Aging Time 300 sec
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Gi0/0	Root	FWD	4	128.1	Shr
Gi0/1	Altn	BLK	4	128.2	Shr
Gi0/2	Altn	BLK	4	128.3	Shr
Gi0/3	Altn	BLK	4	128.4	Shr
Gi1/0	Desg	FWD	4	128.5	Shr
Gi1/1	Desg	FWD	4	128.6	Shr

Layer2Switch-4:

```
vIOS-L2-01>show spanning-tree vlan 1
```

```
VLAN0001
```

```
Spanning tree enabled protocol ieee
```

```
Root ID    Priority    4097
           Address    0c3a.993f.0000
           Cost      4
           Port      1 (GigabitEthernet0/0)
           Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
```

```
Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
           Address    0c7e.ec18.0000
           Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
           Aging Time 300 sec
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Gi0/0	Root	FWD	4	128.1	Shr
Gi0/1	Altn	BLK	4	128.2	Shr
Gi0/2	Altn	BLK	4	128.3	Shr
Gi0/3	Altn	BLK	4	128.4	Shr
Gi1/0	Desg	FWD	4	128.5	Shr
Gi1/1	Desg	FWD	4	128.6	Shr

Layer2Switch-5:

```
vIOS-L2-01>show spanning-tree vlan 1
```

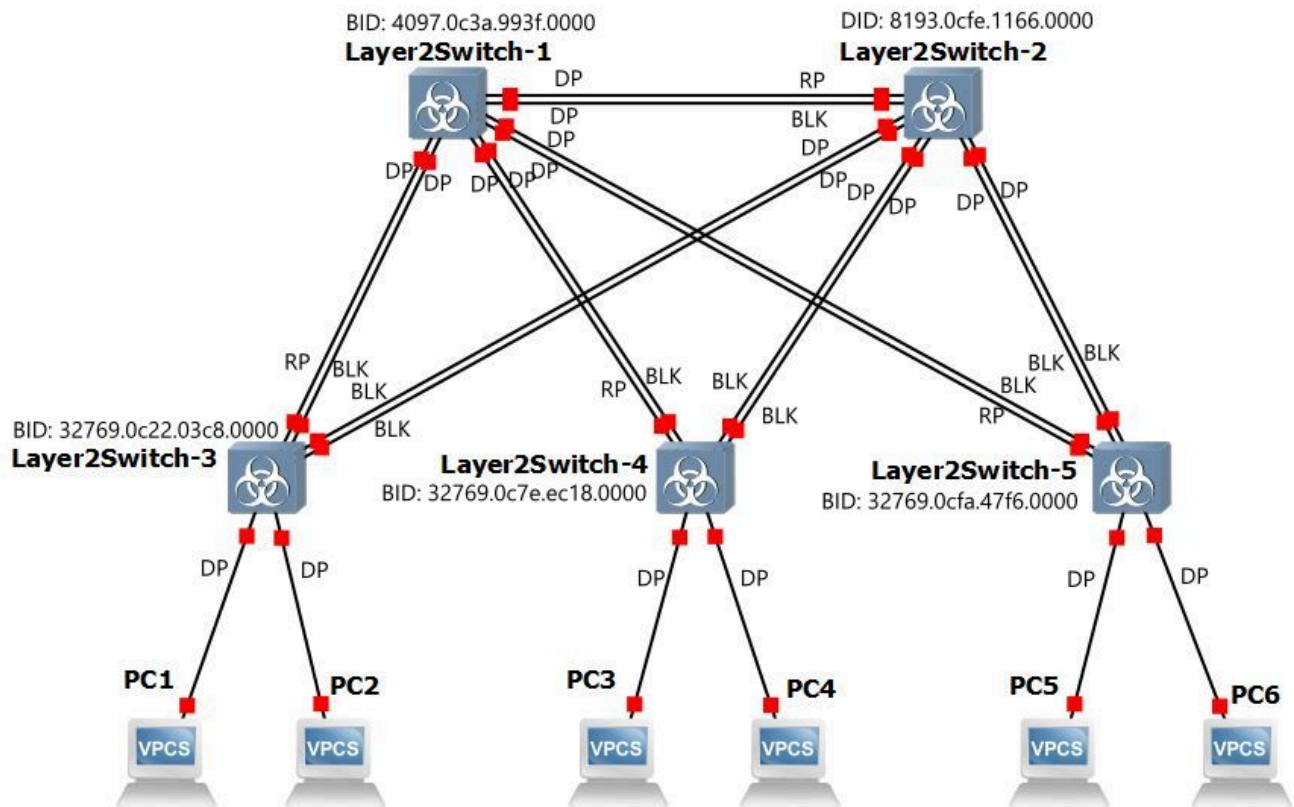
VLAN0001

Spanning tree enabled protocol ieee

Root ID Priority 4097
Address 0c3a.993f.0000
Cost 4
Port 1 (GigabitEthernet0/0)
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)
Address 0cfa.47f6.0000
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.Nbr	Type
Gi0/0	Root	FWD	4	128.1	Shr
Gi0/1	Altn	BLK	4	128.2	Shr
Gi0/2	Altn	BLK	4	128.3	Shr
Gi0/3	Altn	BLK	4	128.4	Shr
Gi1/0	Desg	FWD	4	128.5	Shr
Gi1/1	Desg	FWD	4	128.6	Shr



4. Отслеживание пакетов Hello

Для захвата пакетов Hello BPDU в Wireshark, выбираем линк, на котором анализируем трафик, и в фильтрах Wireshark вводим:

```
stp.type == 0x00
```

Layer2Switch-1

Layer2Switch-2

Захват с Standard input [Layer2Switch-1 Ethernet1 to Layer2Switch-2 Ethernet1]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

stp.type == 0x00

No.	Time	Source	Destination	Protocol	Leng	Info
1076	294.791930	0c:3a:99:3f:00:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8002
1084	297.358091	0c:3a:99:3f:00:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8002
1092	299.891148	0c:3a:99:3f:00:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8002
1100	302.325488	0c:3a:99:3f:00:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8002
1111	304.876649	0c:3a:99:3f:00:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8002
1121	307.468180	0c:3a:99:3f:00:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8002
1129	309.828835	0c:3a:99:3f:00:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8002
1137	312.209120	0c:3a:99:3f:00:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8002
1145	314.798440	0c:3a:99:3f:00:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8002
1155	317.247033	0c:3a:99:3f:00:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8002
1164	319.760060	0c:3a:99:3f:00:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8002
1172	322.216760	0c:3a:99:3f:00:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8002
1181	324.574706	0c:3a:99:3f:00:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8002
1189	327.203007	0c:3a:99:3f:00:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8002
1198	329.789609	0c:3a:99:3f:00:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8002
1208	332.253201	0c:3a:99:3f:00:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8002
1215	334.625585	0c:3a:99:3f:00:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8002
1223	336.923215	0c:3a:99:3f:00:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8002
1231	339.353345	0c:3a:99:3f:00:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8002
1240	341.731283	0c:3a:99:3f:00:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8002
1250	344.158392	0c:3a:99:3f:00:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8002
1259	346.443972	0c:3a:99:3f:00:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8002
1266	348.807037	0c:3a:99:3f:00:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8002
1273	351.086308	0c:3a:99:3f:00:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8002
1285	353.471393	0c:3a:99:3f:00:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8002
1294	355.955519	0c:3a:99:3f:00:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8002
1304	358.539723	0c:3a:99:3f:00:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8002
1313	361.014185	0c:3a:99:3f:00:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8002

Frame 1198: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0

Ethernet II, Src: 0c:3a:99:3f:00:01 (0c:3a:99:3f:00:01), Dst: Nearest-Customer-Bridge (01:80:c2:00:00:00)

802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 300

Logical-Link Control

Spanning Tree Protocol

Protocol Identifier: Spanning Tree Protocol (0x0000)

Protocol Version Identifier: Spanning Tree (0)

BPDU Type: Configuration (0x00)

BPDU flags: 0x00

Root Identifier: 32768 / 300 / 0c:3a:99:3f:00:00

Root Path Cost: 0

Bridge Identifier: 32768 / 300 / 0c:3a:99:3f:00:00

Port identifier: 0x8002

Message Age: 0

Max Age: 20

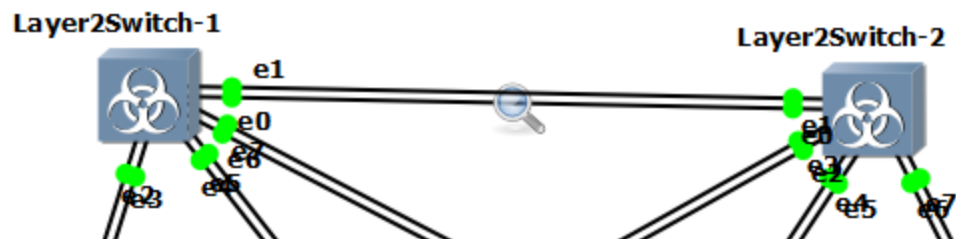
Hello Time: 2

Forward Delay: 15

Тип пакета указан Hello:

```
Protocol Version Identifier: Spanning Tree (0)
BPDU Type: Configuration (0x00)
BPDU flags: 0x00
Root Identifier: 32768 / 300 / 0c:3a:99:3f:00:00
```

Далее, для остальных линков без комментириев:



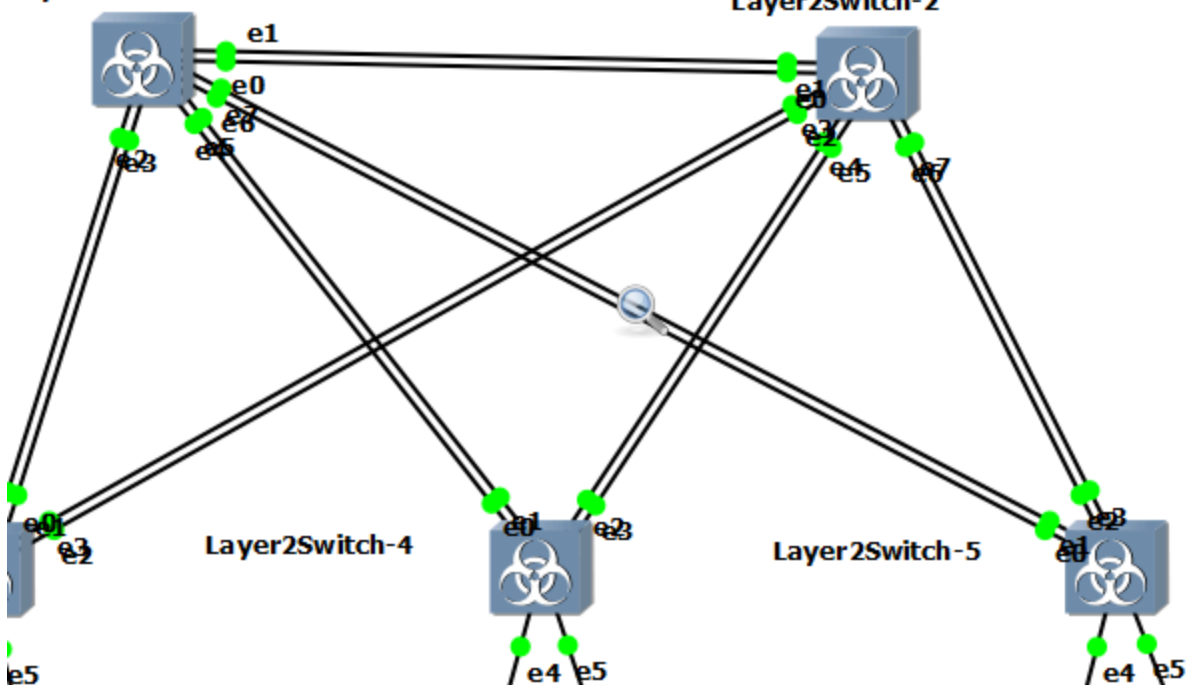
Захват с Standard input [Layer2Switch-1 Ethernet0 to Layer2Switch-2 Ethernet0]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

stp.type == 0x00

No.	Time	Source	Destination	Protocol	Leng	Info
69	19.993181	0c:fe:11:66:00:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:fe:11:66:00:00 Cost = 0 Port = 0x8001
71	20.085549	0c:3a:99:3f:00:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8001
72	20.126109	0c:fe:11:66:00:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:fe:11:66:00:00 Cost = 0 Port = 0x8001
73	20.131139	0c:3a:99:3f:00:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8001
74	20.204133	0c:fe:11:66:00:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:fe:11:66:00:00 Cost = 0 Port = 0x8001
75	20.218483	0c:3a:99:3f:00:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8001
76	20.491839	0c:3a:99:3f:00:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8001
77	21.408505	0c:3a:99:3f:00:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8001
78	22.157811	0c:fe:11:66:00:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:fe:11:66:00:00 Cost = 0 Port = 0x8001
79	22.302831	0c:3a:99:3f:00:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8001
80	22.378820	0c:fe:11:66:00:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:fe:11:66:00:00 Cost = 0 Port = 0x8001
81	22.632248	0c:3a:99:3f:00:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8001
82	22.669121	0c:3a:99:3f:00:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8001
83	22.733000	0c:3a:99:3f:00:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8001
84	22.957176	0c:3a:99:3f:00:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8001

Layer2Switch-1

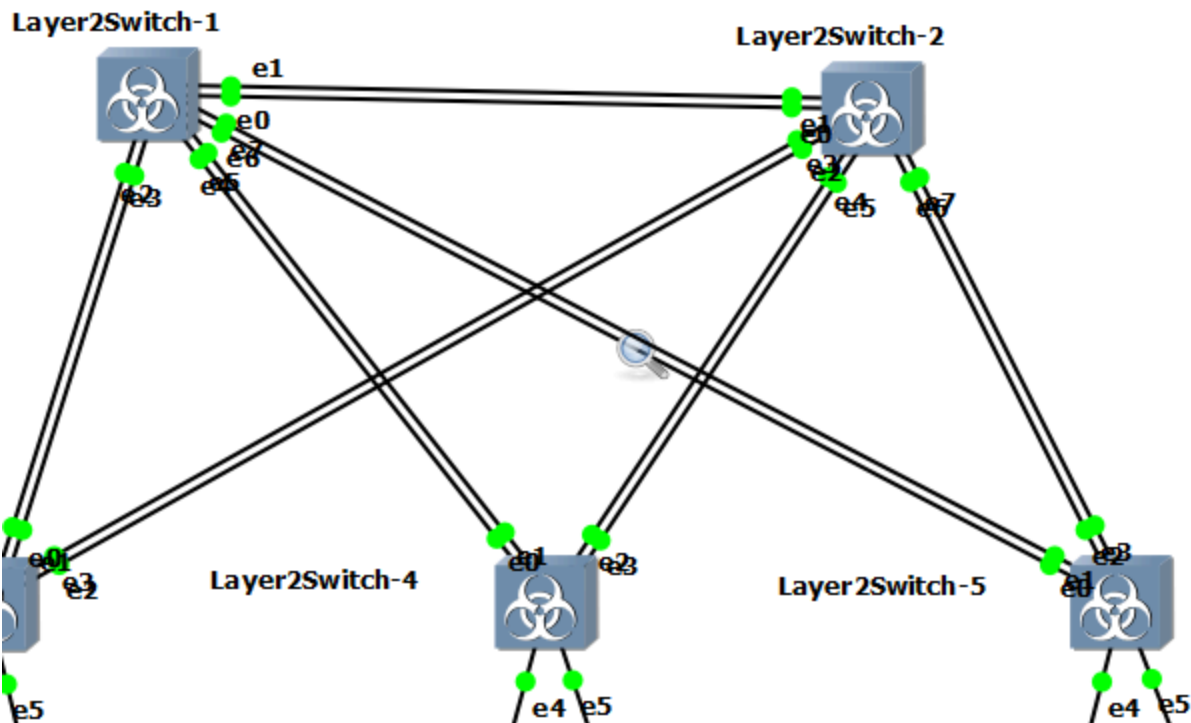


Захват с Standard input [Layer2Switch-1 Ethernet7 to Layer2Switch-5 Ethernet1]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

stp.type == 0x00

No.	Time	Source	Destination	Protocol	Leng	Info
22	4.610417	0c:3a:99:3f:00:07	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8008
23	5.506643	0c:3a:99:3f:00:07	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8008
24	5.581376	0c:3a:99:3f:00:07	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8008
25	5.819358	0c:3a:99:3f:00:07	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8008
26	5.823327	0c:3a:99:3f:00:07	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8008
27	6.151095	0c:fa:47:f6:00:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:fa:47:f6:00:00 Cost = 0 Port = 0x8002
28	6.152357	0c:fa:47:f6:00:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:fa:47:f6:00:00 Cost = 0 Port = 0x8002
29	6.162248	0c:fa:47:f6:00:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:fa:47:f6:00:00 Cost = 0 Port = 0x8002
30	7.058037	0c:3a:99:3f:00:07	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8008
31	8.005592	0c:3a:99:3f:00:07	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8008
32	8.080665	0c:3a:99:3f:00:07	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8008
33	8.309694	0c:fa:47:f6:00:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:fa:47:f6:00:00 Cost = 0 Port = 0x8002
34	8.310900	0c:fa:47:f6:00:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:fa:47:f6:00:00 Cost = 0 Port = 0x8002
35	8.320767	0c:fa:47:f6:00:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:fa:47:f6:00:00 Cost = 0 Port = 0x8002
36	8.340025	0c:3a:99:3f:00:07	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8008
37	8.344464	0c:3a:99:3f:00:07	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8008
38	10.372852	0c:fa:47:f6:00:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:fa:47:f6:00:00 Cost = 0 Port = 0x8002
39	10.373616	0c:fa:47:f6:00:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:fa:47:f6:00:00 Cost = 0 Port = 0x8002



Захват с Standard input [Layer2Switch-1 Ethernet6 to Layer2Switch-5 Ethernet0]

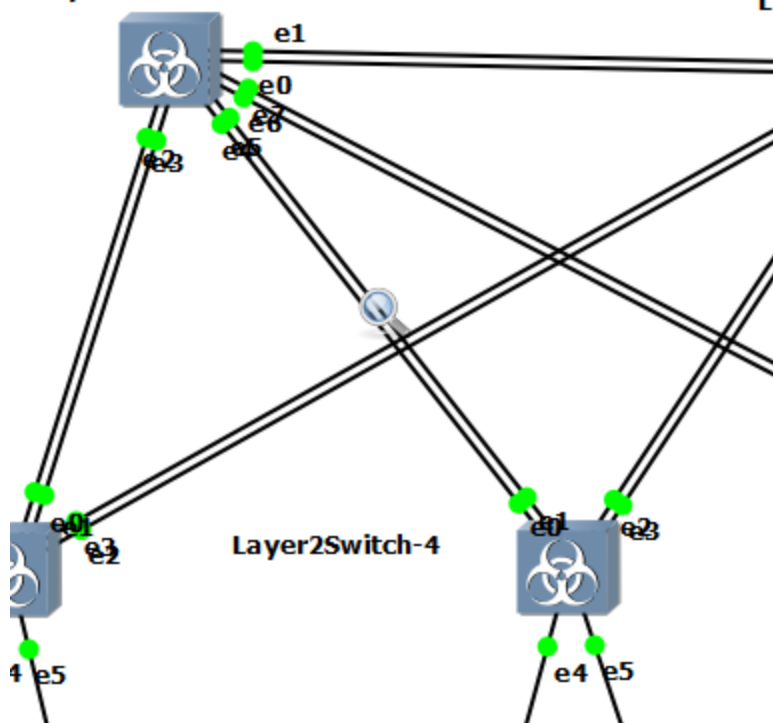
Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

stp.type == 0x00

No.	Time	Source	Destination	Protocol	Leng	Info
22	4.466427	0c:fa:47:f6:00:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:fa:47:f6:00:00 Cost = 0 Port = 0x8001
23	5.385709	0c:3a:99:3f:00:06	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8007
24	6.244572	0c:3a:99:3f:00:06	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8007
25	6.337337	0c:3a:99:3f:00:06	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8007
26	6.498667	0c:fa:47:f6:00:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:fa:47:f6:00:00 Cost = 0 Port = 0x8001
27	6.500585	0c:fa:47:f6:00:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:fa:47:f6:00:00 Cost = 0 Port = 0x8001
28	6.516444	0c:fa:47:f6:00:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:fa:47:f6:00:00 Cost = 0 Port = 0x8001
29	6.600811	0c:3a:99:3f:00:06	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8007
30	6.601759	0c:3a:99:3f:00:06	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8007
31	8.627170	0c:fa:47:f6:00:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:fa:47:f6:00:00 Cost = 0 Port = 0x8001
32	8.627913	0c:3a:99:3f:00:06	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8007
33	8.628300	0c:fa:47:f6:00:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:fa:47:f6:00:00 Cost = 0 Port = 0x8001
34	8.640836	0c:fa:47:f6:00:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:fa:47:f6:00:00 Cost = 0 Port = 0x8001
35	8.800985	0c:3a:99:3f:00:06	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8007
36	8.873718	0c:3a:99:3f:00:06	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8007
37	9.078147	0c:3a:99:3f:00:06	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8007
38	9.437118	0c:3a:99:3f:00:06	0c:3a:99:3f:00:06	LOOP	60	Reply
39	9.833855	0c:3a:99:3f:00:06	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8007
40	10.750713	0c:fa:47:f6:00:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:fa:47:f6:00:00 Cost = 0 Port = 0x8001
41	10.751783	0c:fa:47:f6:00:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:fa:47:f6:00:00 Cost = 0 Port = 0x8001
42	10.765567	0c:fa:47:f6:00:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:fa:47:f6:00:00 Cost = 0 Port = 0x8001

Layer2Switch-1

L

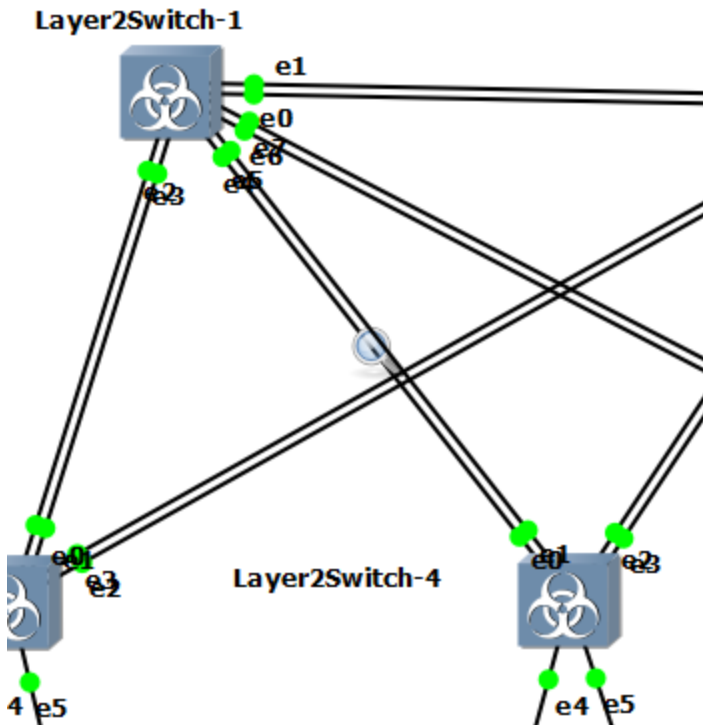


Захват с Standard input [Layer2Switch-1 Ethernet5 to Layer2Switch-4 Ethernet1]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

stp.type == 0x00

No.	Time	Source	Destination	Protocol	Leng	Info
15	2.941151	0c:7e:ec:18:00:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:7e:ec:18:00:00 Cost = 0 Port = 0x8002
16	4.034636	0c:3a:99:3f:00:05	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8006
17	4.941910	0c:7e:ec:18:00:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:7e:ec:18:00:00 Cost = 0 Port = 0x8002
18	4.950928	0c:7e:ec:18:00:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:7e:ec:18:00:00 Cost = 0 Port = 0x8002
19	5.039160	0c:3a:99:3f:00:05	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8006
20	5.045166	0c:7e:ec:18:00:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:7e:ec:18:00:00 Cost = 0 Port = 0x8002
21	5.118184	0c:3a:99:3f:00:05	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8006
22	5.262473	0c:3a:99:3f:00:05	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8006
23	5.331775	0c:3a:99:3f:00:05	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8006
24	5.640441	0c:3a:99:3f:00:05	0c:3a:99:3f:00:05	LOOP	60	Reply
25	6.477692	0c:3a:99:3f:00:05	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8006
26	7.115605	0c:7e:ec:18:00:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:7e:ec:18:00:00 Cost = 0 Port = 0x8002
27	7.124871	0c:7e:ec:18:00:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:7e:ec:18:00:00 Cost = 0 Port = 0x8002
28	7.224890	0c:7e:ec:18:00:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:7e:ec:18:00:00 Cost = 0 Port = 0x8002
29	7.451164	0c:3a:99:3f:00:05	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8006
30	7.532651	0c:3a:99:3f:00:05	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8006
31	7.671199	0c:3a:99:3f:00:05	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8006
32	7.745852	0c:3a:99:3f:00:05	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8006



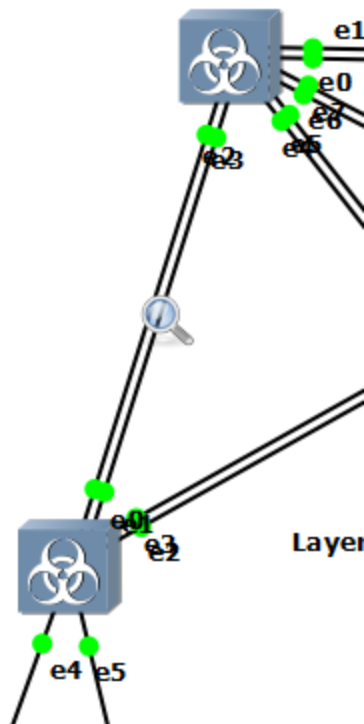
Захват с Standard input [Layer2Switch-1 Ethernet4 to Layer2Switch-4 Ethernet0]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

stp.type == 0x00

No.	Time	Source	Destination	Protocol	Leng	Info
19	4.416160	0c:7e:ec:18:00:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:7e:ec:18:00:00 Cost = 0 Port = 0x8001
20	4.424160	0c:7e:ec:18:00:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:7e:ec:18:00:00 Cost = 0 Port = 0x8001
21	4.519331	0c:7e:ec:18:00:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:7e:ec:18:00:00 Cost = 0 Port = 0x8001
22	5.168245	0c:3a:99:3f:00:04	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8005
23	5.262697	0c:3a:99:3f:00:04	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8005
24	5.547822	0c:3a:99:3f:00:04	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8005
25	5.555273	0c:3a:99:3f:00:04	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8005
26	6.492850	0c:7e:ec:18:00:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:7e:ec:18:00:00 Cost = 0 Port = 0x8001
27	6.499015	0c:7e:ec:18:00:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:7e:ec:18:00:00 Cost = 0 Port = 0x8001
28	6.595004	0c:7e:ec:18:00:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:7e:ec:18:00:00 Cost = 0 Port = 0x8001
29	6.750100	0c:3a:99:3f:00:04	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8005
30	7.652992	0c:3a:99:3f:00:04	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8005
31	7.743909	0c:3a:99:3f:00:04	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8005
32	7.953785	0c:3a:99:3f:00:04	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8005
33	7.954072	0c:3a:99:3f:00:04	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8005
34	8.719547	0c:7e:ec:18:00:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:7e:ec:18:00:00 Cost = 0 Port = 0x8001
35	8.725781	0c:7e:ec:18:00:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:7e:ec:18:00:00 Cost = 0 Port = 0x8001
36	8.822195	0c:7e:ec:18:00:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:7e:ec:18:00:00 Cost = 0 Port = 0x8001

Layer2Switch-1



Захват с Standard input [Layer2Switch-1 Ethernet3 to Layer2Switch-3 Ethernet1]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

stp.type == 0x00

No.	Time	Source	Destination	Protocol	Leng	Info
18	3.581889	0c:fe:11:66:00:01	DEC-MOP-Remote-Console	0x6002	77	DEC DNA Remote Console
19	3.947828	0c:3a:99:3f:00:03	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8004
20	4.830414	0c:3a:99:3f:00:03	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8004
21	4.918810	0c:3a:99:3f:00:03	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8004
22	5.033932	0c:22:03:c8:00:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:22:03:c8:00:00 Cost = 0 Port = 0x8002
23	5.085714	0c:22:03:c8:00:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:22:03:c8:00:00 Cost = 0 Port = 0x8002
24	5.098674	0c:22:03:c8:00:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:22:03:c8:00:00 Cost = 0 Port = 0x8002
25	5.163871	0c:3a:99:3f:00:03	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8004
26	5.164202	0c:3a:99:3f:00:03	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8004
27	6.034720	0c:22:03:c8:00:01	0c:22:03:c8:00:01	LOOP	60	Reply
28	6.309072	0c:3a:99:3f:00:03	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8004
29	7.265562	0c:22:03:c8:00:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:22:03:c8:00:00 Cost = 0 Port = 0x8002
30	7.317509	0c:22:03:c8:00:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:22:03:c8:00:00 Cost = 0 Port = 0x8002
31	7.330611	0c:22:03:c8:00:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:22:03:c8:00:00 Cost = 0 Port = 0x8002

Layer2Switch-1



Захват с Standard input [Layer2Switch-1 Ethernet2 to Layer2Switch-3 Ethernet0]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

stp.type == 0x00

No.	Time	Source	Destination	Protocol	Leng	Info
19	5.067610	0c:22:03:c8:00:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:22:03:c8:00:00 Cost = 0 Port = 0x8001
20	5.077271	0c:22:03:c8:00:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:22:03:c8:00:00 Cost = 0 Port = 0x8001
21	5.171614	0c:3a:99:3f:00:02	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8003
22	5.226731	0c:3a:99:3f:00:02	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8003
23	5.236084	0c:3a:99:3f:00:02	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8003
24	5.508392	0c:3a:99:3f:00:02	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8003
25	5.794693	0c:3a:99:3f:00:02	0c:3a:99:3f:00:02	LOOP	60	Reply
26	5.900223	0c:22:03:c8:00:00	0c:22:03:c8:00:00	LOOP	60	Reply
27	6.463280	0c:3a:99:3f:00:02	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8003
28	7.046841	0c:22:03:c8:00:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:22:03:c8:00:00 Cost = 0 Port = 0x8001
29	7.098667	0c:22:03:c8:00:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:22:03:c8:00:00 Cost = 0 Port = 0x8001
30	7.109004	0c:22:03:c8:00:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:22:03:c8:00:00 Cost = 0 Port = 0x8001
31	7.631288	0c:3a:99:3f:00:02	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8003
32	7.684992	0c:3a:99:3f:00:02	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8003
33	7.695150	0c:3a:99:3f:00:02	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8003
34	7.896213	0c:3a:99:3f:00:02	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8003
35	8.843830	0c:3a:99:3f:00:02	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:3a:99:3f:00:00 Cost = 0 Port = 0x8003
36	9.084488	0c:22:03:c8:00:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:22:03:c8:00:00 Cost = 0 Port = 0x8001
37	9.139282	0c:22:03:c8:00:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:22:03:c8:00:00 Cost = 0 Port = 0x8001
38	9.148698	0c:22:03:c8:00:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:22:03:c8:00:00 Cost = 0 Port = 0x8001

Diagram showing a network topology with five Layer2Switches (Layer2Switch-1, Layer2Switch-2, Layer2Switch-3, Layer2Switch-4, Layer2Switch-5) connected via Ethernet ports. The switches are interconnected in a mesh-like structure.

Below the diagram is a packet capture window titled "Захват из - [Layer2Switch-2 Ethernet6 to Layer2Switch-5 Ethernet2]". The window displays a list of captured packets and a detailed view of the selected packet (Frame 1).

Packet List:

No.	Time	Source	Destination	Protocol	Length	Info
34	8.825307	0c:fe:11:66:00:06	Spanning-tree-(for...	STP	60	Conf. Root = 32768/200/0c:fe:11:66:00:00 Cost = 0 Port = 0x...
35	8.868291	0c:fa:47:f6:00:02	0c:fa:47:f6:00:02	LOOP	60	Reply
36	9.369099	0c:fa:47:f6:00:02	Spanning-tree-(for...	STP	60	Conf. Root = 32768/100/0c:fa:47:f6:00:00 Cost = 0 Port = 0x...
37	9.517094	0c:fa:47:f6:00:02	Spanning-tree-(for...	STP	60	Conf. Root = 32768/200/0c:fa:47:f6:00:00 Cost = 0 Port = 0x...
38	9.577571	0c:fe:11:66:00:06	Spanning-tree-(for...	STP	60	Conf. Root = 4096/1/0c:3a:99:3f:00:00 Cost = 4 Port = 0x8007
39	9.604085	0c:fa:47:f6:00:02	Spanning-tree-(for...	STP	60	Conf. Root = 32768/300/0c:fa:47:f6:00:00 Cost = 0 Port = 0x...
40	9.739811	0c:fe:11:66:00:06	Spanning-tree-(for...	STP	60	Conf. Root = 32768/300/0c:fe:11:66:00:00 Cost = 0 Port = 0x...
41	10.471433	0c:fe:11:66:00:06	Spanning-tree-(for...	STP	60	Conf. Root = 32768/100/0c:fe:11:66:00:00 Cost = 0 Port = 0x...
42	10.622172	0c:fe:11:66:00:06	Spanning-tree-(for...	STP	60	Conf. Root = 4096/1/0c:3a:99:3f:00:00 Cost = 4 Port = 0x8007
43	10.981401	0c:fe:11:66:00:06	Spanning-tree-(for...	STP	60	Conf. Root = 32768/200/0c:fe:11:66:00:00 Cost = 0 Port = 0x...
44	11.372820	0c:fa:47:f6:00:02	Spanning-tree-(for...	STP	60	Conf. Root = 32768/100/0c:fa:47:f6:00:00 Cost = 0 Port = 0x...

Frame 1 Details:

Frame 1: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on 0000
Ethernet II, Src: 0c:fe:11:66:00:06 (0c:fe:11:66:00:06), Dst: Spanning-tree-(for...
802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 200
Logical-Link Control
Spanning Tree Protocol

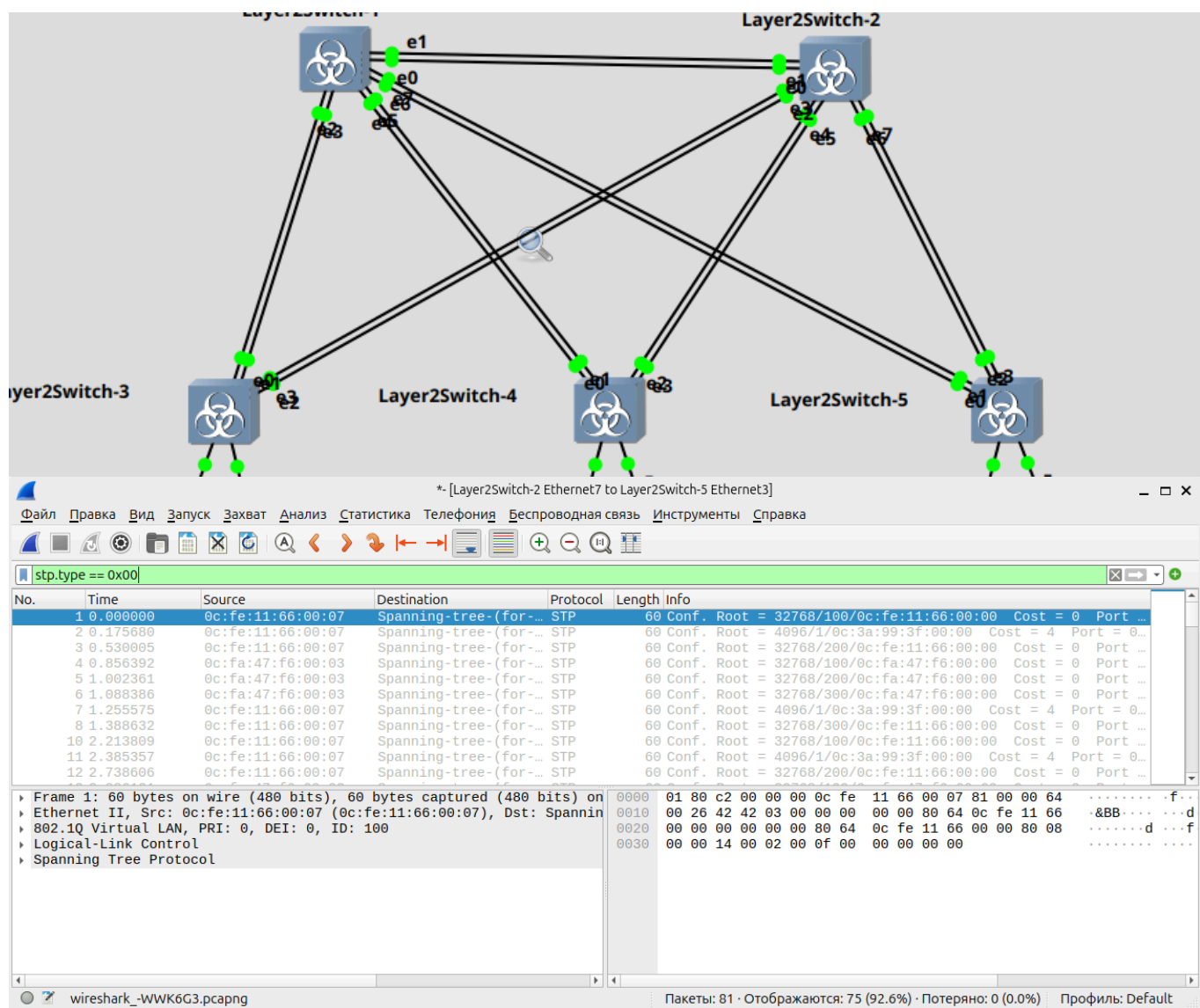
Hex dump and ASCII representation of the frame data:

Hex	ASCII
0000 01 80 c2 00 00 00 0c fe 11 66 00 06 81 00 00 c8	f..
0010 00 26 42 42 03 00 00 00 00 00 80 c8 0c fe 11 66	..&BB.....
0020 00 00 00 00 00 00 80 c8 0c fe 11 66 00 00 80 07	f..
0030 00 00 14 00 02 00 0f 00 00 00 00 00	

Spanning Tree Protocol: Protocol

Пакеты: 44 · Отображаются: 44 (100.0%)

Профиль: Default



Захват из - [Layer2Switch-2 Ethernet6 to Layer2Switch-5 Ethernet2]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

stp.type == 0x00

No.	Time	Source	Destination	Protocol	Length	Info
34	8.825307	0c:fe:11:66:00:06	Spanning-tree-(for...	STP	60	Conf. Root = 32768/200/0c:fe:11:66:00:00 Cost = 0 Port = 0x...
35	8.868291	0c:fa:47:f6:00:02	0c:fa:47:f6:00:02	LOOP	60	Reply
36	9.369099	0c:fa:47:f6:00:02	Spanning-tree-(for...	STP	60	Conf. Root = 32768/100/0c:fa:47:f6:00:00 Cost = 0 Port = 0x...
37	9.517094	0c:fa:47:f6:00:02	Spanning-tree-(for...	STP	60	Conf. Root = 32768/200/0c:fa:47:f6:00:00 Cost = 0 Port = 0x...
38	9.577571	0c:fe:11:66:00:06	Spanning-tree-(for...	STP	60	Conf. Root = 4096/1/0c:3a:99:3f:00:00 Cost = 4 Port = 0x8007
39	9.604085	0c:fa:47:f6:00:02	Spanning-tree-(for...	STP	60	Conf. Root = 32768/300/0c:fa:47:f6:00:00 Cost = 0 Port = 0x...
40	9.739811	0c:fe:11:66:00:06	Spanning-tree-(for...	STP	60	Conf. Root = 32768/300/0c:fe:11:66:00:00 Cost = 0 Port = 0x...
41	10.471433	0c:fe:11:66:00:06	Spanning-tree-(for...	STP	60	Conf. Root = 32768/100/0c:fe:11:66:00:00 Cost = 0 Port = 0x...
42	10.622172	0c:fe:11:66:00:06	Spanning-tree-(for...	STP	60	Conf. Root = 4096/1/0c:3a:99:3f:00:00 Cost = 4 Port = 0x8007
43	10.981401	0c:fe:11:66:00:06	Spanning-tree-(for...	STP	60	Conf. Root = 32768/200/0c:fe:11:66:00:00 Cost = 0 Port = 0x...
44	11.372820	0c:fa:47:f6:00:02	Spanning-tree-(for...	STP	60	Conf. Root = 32768/100/0c:fa:47:f6:00:00 Cost = 0 Port = 0x...

Frame 1: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on
 Ethernet II, Src: 0c:fe:11:66:00:06 (0c:fe:11:66:00:06), Dst: Spanning
 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 200
 Logical-Link Control
 Spanning Tree Protocol

Spanning Tree Protocol: Protocol

Пакеты: 44 · Отображаются: 44 (100.0%)

Профиль: Default

Layer2Switch-1

Layer2Switch-2

Layer2Switch-3

Layer2Switch-4

Layer2Switch-5

*- [Layer2Switch-2 Ethernet7 to Layer2Switch-5 Ethernet3]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

stp.type == 0x00

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	0c:fe:11:66:00:07	Spanning-tree-(for...	STP	60	Conf. Root = 32768/100/0c:fe:11:66:00:00 Cost = 0 Port ...
2	0.175680	0c:fe:11:66:00:07	Spanning-tree-(for...	STP	60	Conf. Root = 4096/1/0c:3a:99:3f:00:00 Cost = 4 Port = 0...
3	0.530005	0c:fe:11:66:00:07	Spanning-tree-(for...	STP	60	Conf. Root = 32768/200/0c:fe:11:66:00:00 Cost = 0 Port ...
4	0.856392	0c:fa:47:f6:00:03	Spanning-tree-(for...	STP	60	Conf. Root = 32768/100/0c:fa:47:f6:00:00 Cost = 0 Port ...
5	1.002361	0c:fa:47:f6:00:03	Spanning-tree-(for...	STP	60	Conf. Root = 32768/200/0c:fa:47:f6:00:00 Cost = 0 Port ...
6	1.088386	0c:fa:47:f6:00:03	Spanning-tree-(for...	STP	60	Conf. Root = 32768/300/0c:fa:47:f6:00:00 Cost = 0 Port ...
7	1.255575	0c:fe:11:66:00:07	Spanning-tree-(for...	STP	60	Conf. Root = 4096/1/0c:3a:99:3f:00:00 Cost = 4 Port = 0...
8	1.388632	0c:fe:11:66:00:07	Spanning-tree-(for...	STP	60	Conf. Root = 32768/300/0c:fe:11:66:00:00 Cost = 0 Port ...
10	2.213809	0c:fe:11:66:00:07	Spanning-tree-(for...	STP	60	Conf. Root = 32768/100/0c:fe:11:66:00:00 Cost = 0 Port ...
11	2.385357	0c:fe:11:66:00:07	Spanning-tree-(for...	STP	60	Conf. Root = 4096/1/0c:3a:99:3f:00:00 Cost = 4 Port = 0...
12	2.738606	0c:fe:11:66:00:07	Spanning-tree-(for...	STP	60	Conf. Root = 32768/200/0c:fe:11:66:00:00 Cost = 0 Port ...

Frame 1: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on
 Ethernet II, Src: 0c:fe:11:66:00:07 (0c:fe:11:66:00:07), Dst: Spanning
 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 100
 Logical-Link Control
 Spanning Tree Protocol

0000 01 80 c2 00 00 00 0c fe 11 66 00 07 81 00 00 64f..
 0010 00 26 42 42 03 00 00 00 00 00 80 64 0c fe 11 66 ..&BB....d
 0020 00 00 00 00 00 00 80 64 0c fe 11 66 00 00 80 08d...f
 0030 00 00 14 00 02 00 0f 00 00 00 00 00

wireshark_ - WWK6G3.pcapng

Пакеты: 81 · Отображаются: 75 (92.6%) · Потеряно: 0 (0.0%) · Профиль: Default

Захват из - [Layer2Switch-2 Ethernet6 to Layer2Switch-5 Ethernet2]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

stp.type == 0x00

No.	Time	Source	Destination	Protocol	Length	Info
34	8.825307	0c:fe:11:66:00:06	Spanning-tree-(for...	STP	60	Conf. Root = 32768/200/0c:fe:11:66:00:00 Cost = 0 Port = 0x...
35	8.868291	0c:fa:47:f6:00:02	0c:fa:47:f6:00:02	LOOP	60	Reply
36	9.369099	0c:fa:47:f6:00:02	Spanning-tree-(for...	STP	60	Conf. Root = 32768/100/0c:fa:47:f6:00:00 Cost = 0 Port = 0x...
37	9.517094	0c:fa:47:f6:00:02	Spanning-tree-(for...	STP	60	Conf. Root = 32768/200/0c:fa:47:f6:00:00 Cost = 0 Port = 0x...
38	9.577571	0c:fe:11:66:00:06	Spanning-tree-(for...	STP	60	Conf. Root = 4096/1/0c:3a:99:3f:00:00 Cost = 4 Port = 0x8007
39	9.604085	0c:fa:47:f6:00:02	Spanning-tree-(for...	STP	60	Conf. Root = 32768/300/0c:fa:47:f6:00:00 Cost = 0 Port = 0x...
40	9.739811	0c:fe:11:66:00:06	Spanning-tree-(for...	STP	60	Conf. Root = 32768/300/0c:fe:11:66:00:00 Cost = 0 Port = 0x...
41	10.471433	0c:fe:11:66:00:06	Spanning-tree-(for...	STP	60	Conf. Root = 32768/100/0c:fe:11:66:00:00 Cost = 0 Port = 0x...
42	10.622172	0c:fe:11:66:00:06	Spanning-tree-(for...	STP	60	Conf. Root = 4096/1/0c:3a:99:3f:00:00 Cost = 4 Port = 0x8007
43	10.981401	0c:fe:11:66:00:06	Spanning-tree-(for...	STP	60	Conf. Root = 32768/200/0c:fe:11:66:00:00 Cost = 0 Port = 0x...
44	11.372820	0c:fa:47:f6:00:02	Spanning-tree-(for...	STP	60	Conf. Root = 32768/100/0c:fa:47:f6:00:00 Cost = 0 Port = 0x...

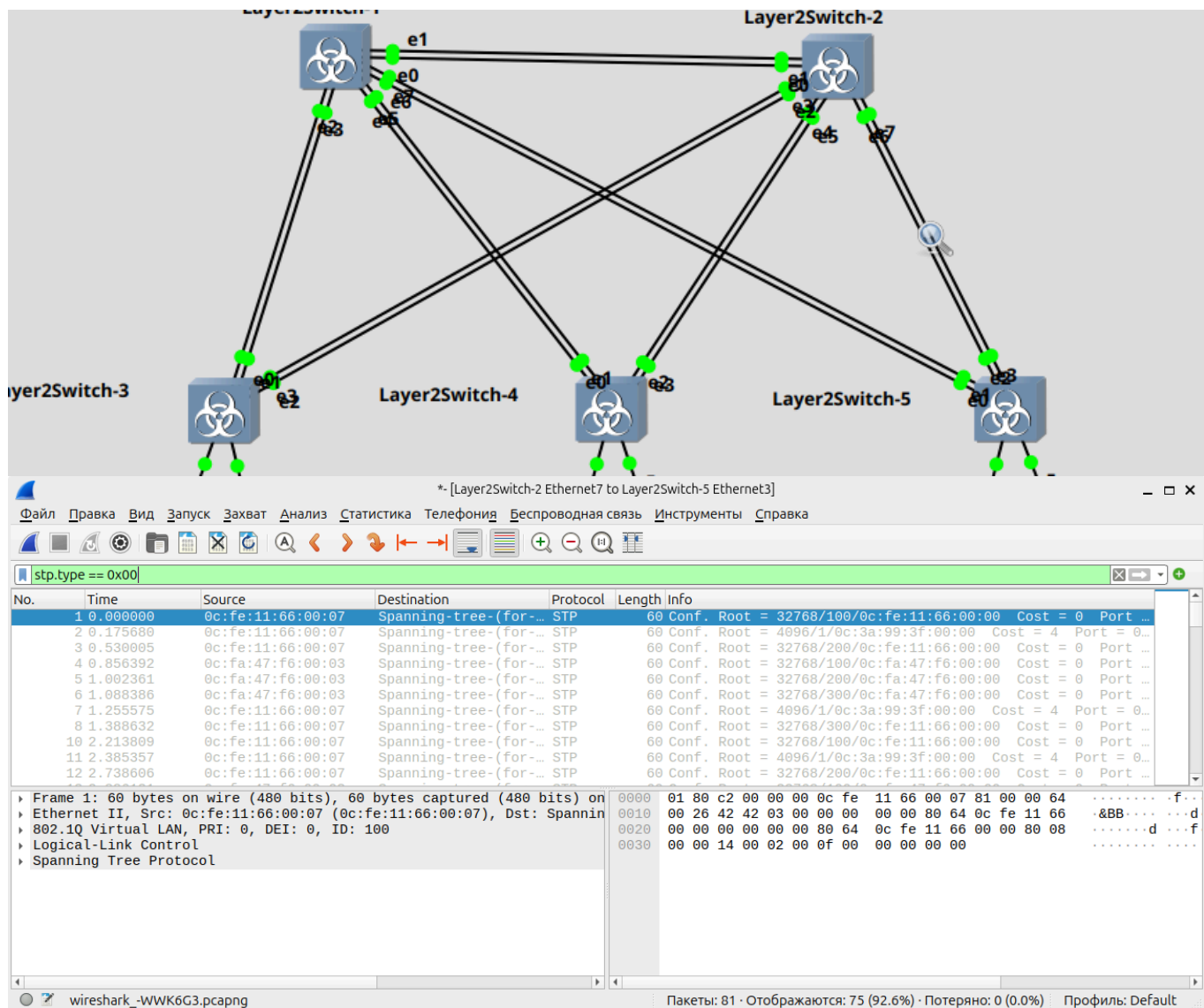
Frame 1: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on 0000
 Ethernet II, Src: 0c:fe:11:66:00:06 (0c:fe:11:66:00:06), Dst: Spanning-tree-(for...
 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 200
 Logical-Link Control
 Spanning Tree Protocol

0000 01 80 c2 00 00 00 0c fe 11 66 00 06 81 00 00 c8f..
 0010 00 26 42 42 03 00 00 00 00 00 80 c8 0c fe 11 66 ..&BB.....
 0020 00 00 00 00 00 00 80 c8 0c fe 11 66 00 00 80 07f..
 0030 00 00 14 00 02 00 0f 00 00 00 00 00f..

Spanning Tree Protocol: Protocol

Пакеты: 44 · Отображаются: 44 (100.0%)

Профиль: Default



5. Изменение стоимости порта коммутатора

Изменим стоимость порта e0 у коммутатора Layer2Switch-3:

```
interface Gi0/0
spanning-tree cost 50
end
write
show spanning-tree vlan 1
```

```

vIOS-L2-01#show spanning-tree vlan 1

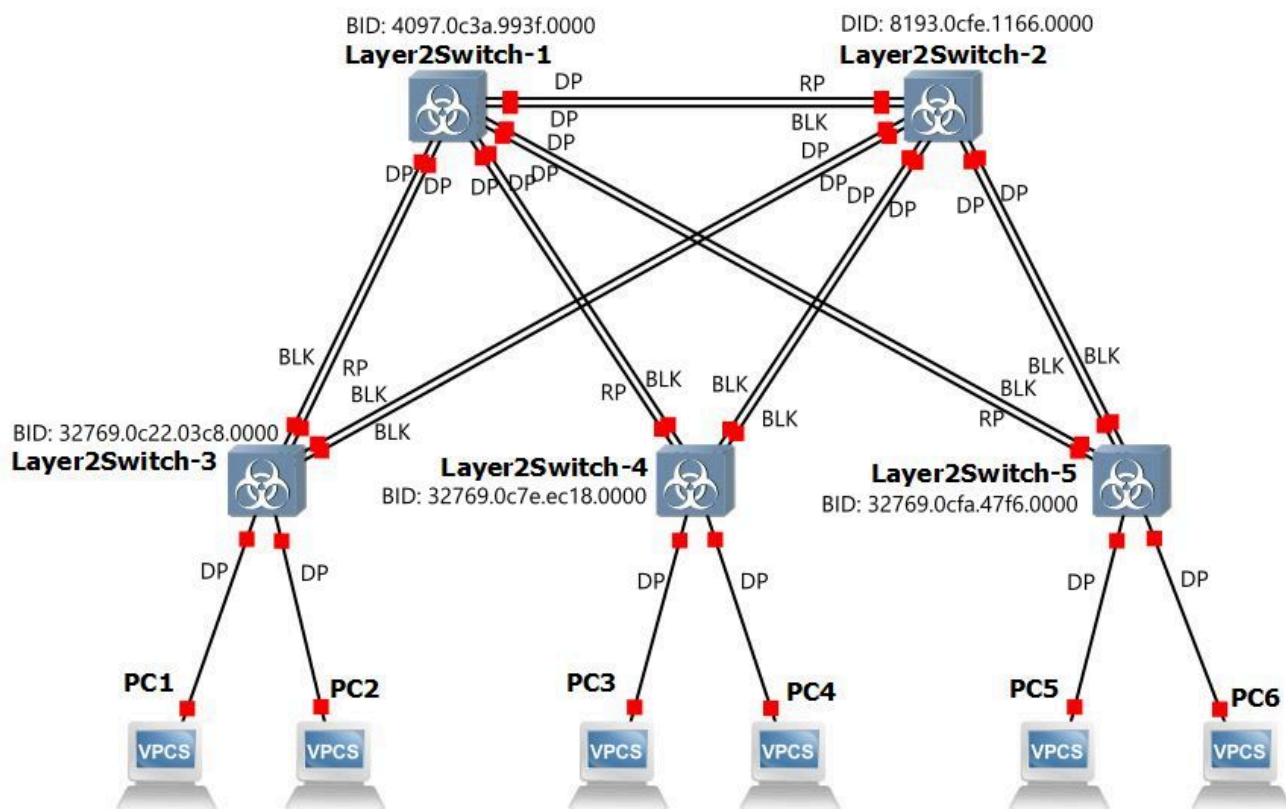
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    4097
            Address     0c3a.993f.0000
            Cost        4
            Port        2 (GigabitEthernet0/1)
            Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
            Address     0c22.03c8.0000
            Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
            Aging Time   15 sec

Interface                Role Sts Cost      Prio.Nbr Type
-----
Gi0/0                    Altn BLK 10       128.1   Shr
Gi0/1                    Root FWD 4        128.2   Shr
Gi0/2                    Altn BLK 4        128.3   Shr
Gi0/3                    Altn BLK 4        128.4   Shr
Gi1/0                    Desg FWD 4        128.5   Shr
Gi1/1                    Desg FWD 4        128.6   Shr

```

Изменения в схеме: у коммутатора Layer2Switch-3 поменялись местами роли у портов Gi0/0 и Gi0/1



6. Файлы конфигурации коммутаторов

Layer2Switch-1.log

===== PuTTY log 2025.12.13 15:12:29 =====

sh run

Building configuration...

Current configuration : 5277 bytes

!

! Last configuration change at 04:45:43 UTC Wed Dec 10 2025

!

version 15.0

service timestamps debug datetime msec

service timestamps log datetime msec

no service password-encryption

service compress-config

!

hostname vIOS-L2-01

!

boot-start-marker

boot-end-marker

!

!

!

no aaa new-model

!

!

!

!

!

vtp domain CISCO-vIOS

vtp mode transparent

!

!

!

ip cef

no ipv6 cef

!

!

!

spanning-tree mode pvst

spanning-tree extend system-id

spanning-tree vlan 1 priority 4096

!

vlan internal allocation policy ascending

!

vlan 100

```
name VLAN100
!
vlan 200,300
!
!
!
!
!
!
!
!
!
!
!
!
interface GigabitEthernet0/0
media-type rj45
negotiation auto
!
interface GigabitEthernet0/1
media-type rj45
negotiation auto
!
interface GigabitEthernet0/2
media-type rj45
negotiation auto
!
interface GigabitEthernet0/3
media-type rj45
negotiation auto
!
interface GigabitEthernet1/0
media-type rj45
negotiation auto
!
interface GigabitEthernet1/1
media-type rj45
negotiation auto
!
```

```
interface GigabitEthernet1/2
media-type rj45
negotiation auto
!
interface GigabitEthernet1/3
media-type rj45
negotiation auto
!
interface GigabitEthernet2/0
media-type rj45
negotiation auto
!
ip forward-protocol nd
!
no ip http server
no ip http secure-server
!
!
!
!
!
!
control-plane
!
banner exec
```

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!

line con 0

logging synchronous

line aux 0

line vty 0 4

logging synchronous

login

line vty 5 15

logging synchronous

login

!

!

end

vIOS-L2-01#

Layer2Switch-2.log

===== PuTTY log 2025.12.13 15:14:39 =====

sh run

Building configuration...

Current configuration : 5277 bytes

!

! Last configuration change at 04:45:43 UTC Wed Dec 10 2025

!

version 15.0

service timestamps debug datetime msec

service timestamps log datetime msec

no service password-encryption

service compress-config

!

hostname vIOS-L2-01

!

boot-start-marker

boot-end-marker

!

!

!

no aaa new-model

!

!

!

!

!

vtp domain CISCO-vIOS

vtp mode transparent

!

!

!

ip cef

no ipv6 cef

!

!

!

spanning-tree mode pvst

spanning-tree extend system-id

spanning-tree vlan 1 priority 8192

!

vlan internal allocation policy ascending

!

vlan 100

name VLAN100

!

vlan 200,300

!
!
!
!
!
!
!
!
!
!
!
!
!

interface GigabitEthernet0/0

media-type rj45

negotiation auto

!

interface GigabitEthernet0/1

media-type rj45

negotiation auto

!

interface GigabitEthernet0/2

media-type rj45

negotiation auto

!

interface GigabitEthernet0/3

media-type rj45

negotiation auto

!

interface GigabitEthernet1/0

media-type rj45

negotiation auto

!

interface GigabitEthernet1/1

media-type rj45

negotiation auto

!

interface GigabitEthernet1/2

media-type rj45

```
negotiation auto
!
interface GigabitEthernet1/3
media-type rj45
negotiation auto
!
interface GigabitEthernet2/0
media-type rj45
negotiation auto
!
ip forward-protocol nd
!
no ip http server
no ip http secure-server
!
!
!
!
!
!
control-plane
!
banner exec
```

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!

line con 0

logging synchronous

line aux 0

line vty 0 4

logging synchronous

login

line vty 5 15

logging synchronous

login

!

!

end

vIOS-L2-01#

Layer2Switch-3.log

===== PuTTY log 2025.12.13 15:15:45 =====

sh run

Building configuration...

Current configuration : 5065 bytes

!

! Last configuration change at 06:39:20 UTC Sat Dec 13 2025

!

version 15.0

```
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
service compress-config
!
hostname vIOS-L2-01
!
boot-start-marker
boot-end-marker
!
!
!
no aaa new-model
!
!
!
!
!
vtp domain CISCO-VIOS
vtp mode transparent
!
!
!
ip cef
no ipv6 cef
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
vlan internal allocation policy ascending
!
vlan 100
name VLAN100
!
vlan 200,300
!
!
!
!
```

```
!  
!  
!  
!  
!  
!  
!  
!  
!  
interface GigabitEthernet0/0  
media-type rj45  
negotiation auto  
spanning-tree cost 10  
!  
interface GigabitEthernet0/1  
media-type rj45  
negotiation auto  
!  
interface GigabitEthernet0/2  
media-type rj45  
negotiation auto  
!  
interface GigabitEthernet0/3  
media-type rj45  
negotiation auto  
!  
interface GigabitEthernet1/0  
media-type rj45  
negotiation auto  
!  
interface GigabitEthernet1/1  
media-type rj45  
negotiation auto  
!  
ip forward-protocol nd  
!  
no ip http server  
no ip http secure-server  
!  
!
```

control-plane

banner exec

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!

line con 0

logging synchronous

line aux 0

```
line vty 0 4
logging synchronous
login
line vty 5 15
logging synchronous
login
!
!
end
```

vIOS-L2-01#

Layer2Switch-4.log

===== PuTTY log 2025.12.13 15:16:34 =====

sh run

Building configuration...

Current configuration : 5042 bytes

!

! Last configuration change at 04:45:44 UTC Wed Dec 10 2025

!

version 15.0

service timestamps debug datetime msec

service timestamps log datetime msec

no service password-encryption

service compress-config

!

hostname vIOS-L2-01

!

boot-start-marker

boot-end-marker

!

!

!

no aaa new-model

!

!

!

```
!  
!  
vtp domain CISCO-VIOS  
vtp mode transparent  
!  
!  
!  
ip cef  
no ipv6 cef  
!  
!  
spanning-tree mode pvst  
spanning-tree extend system-id  
!  
vlan internal allocation policy ascending  
!  
vlan 100  
name VLAN100  
!  
vlan 200,300  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
interface GigabitEthernet0/0  
media-type rj45  
negotiation auto  
!  
interface GigabitEthernet0/1  
media-type rj45  
negotiation auto
```

```
!  
interface GigabitEthernet0/2  
media-type rj45  
negotiation auto  
!  
interface GigabitEthernet0/3  
media-type rj45  
negotiation auto  
!  
interface GigabitEthernet1/0  
media-type rj45  
negotiation auto  
!  
interface GigabitEthernet1/1  
media-type rj45  
negotiation auto  
!  
ip forward-protocol nd  
!  
no ip http server  
no ip http secure-server  
!  
!  
!  
!  
!  
!  
control-plane  
!  
banner exec
```

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!

line con 0

logging synchronous

line aux 0

line vty 0 4

logging synchronous

login

line vty 5 15

logging synchronous

login

!

!

end

vIOS-L2-01#

===== PuTTY log 2025.12.13 15:17:32 =====

sh run

Building configuration...

Current configuration : 5042 bytes

!

! Last configuration change at 04:45:44 UTC Wed Dec 10 2025

!

version 15.0

service timestamps debug datetime msec

service timestamps log datetime msec

no service password-encryption

service compress-config

!

hostname vIOS-L2-01

!

boot-start-marker

boot-end-marker

!

!

!

no aaa new-model

!

!

!

!

!

vtp domain CISCO-vIOS

vtp mode transparent

!

!

!

ip cef

no ipv6 cef

!

!

spanning-tree mode pvst

spanning-tree extend system-id

!

vlan internal allocation policy ascending


```
negotiation auto
!
ip forward-protocol nd
!
no ip http server
no ip http secure-server
!
!
!
!
!
!
control-plane
!
banner exec
```

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!

line con 0

logging synchronous

line aux 0

line vty 0 4

logging synchronous

login

line vty 5 15

logging synchronous

login

!

!

end

vIOS-L2-01#